

Probability-based Learning

Sections 6.1, 6.2, 6.3

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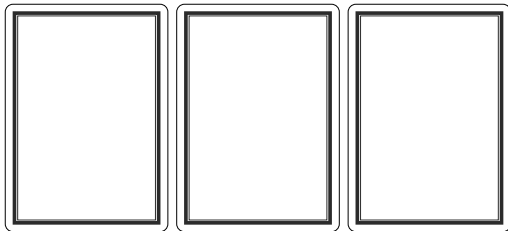
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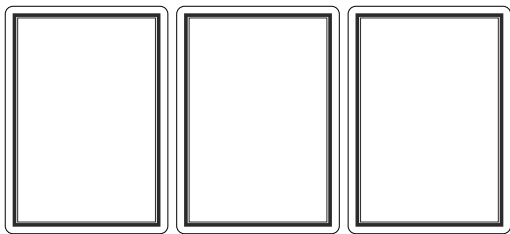


(a)

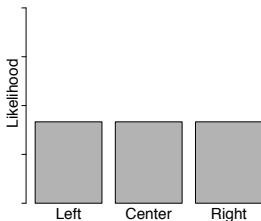


(b)

Figure: A game of *find the lady*



(a)



(b)

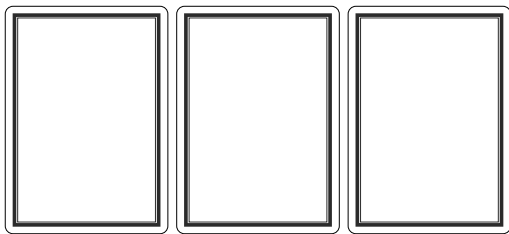
Figure: A game of *find the lady*: (a) the cards dealt face down on a table; and (b) the initial likelihoods of the queen ending up in each position.

Game analysis!!

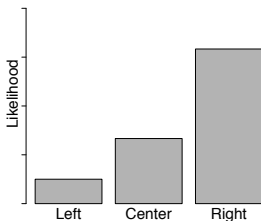
- After observing the dealer play 30 games with other players, The dealer tends to drop the queen in:
 - The right position 19 times.
 - The left position occurs 3 times.
 - The center position 8 times.

Question

How can this evidence impact the initial likelihoods?



(a)



(b)

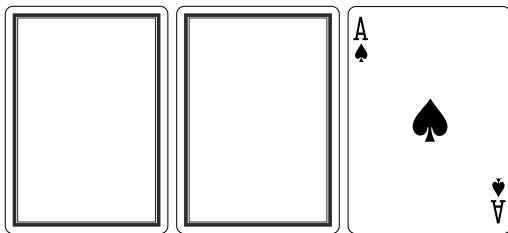
Figure: A game of *find the lady*: (a) the cards dealt face down on a table; and (b) a revised set of likelihoods for the position of the queen based on evidence collected.

You are lucky!!

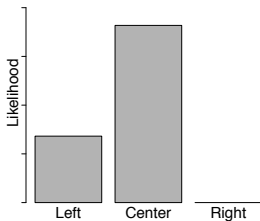
- Sudden gust of wind turns over the card on the right to reveal that it is the ace of spades.

Question

How can this event impact the initial likelihoods?



(a)



(b)

Figure: A game of *find the lady*: (a) The set of cards after the wind blows over the one on the right; (b) the revised likelihoods for the position of the queen based on this new evidence.

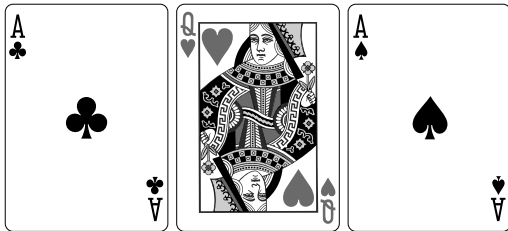


Figure: A game of *find the lady*: The final positions of the cards in the game.

- We can use estimates of likelihoods to determine the most likely prediction that should be made.
- More importantly, we revise these predictions based on data we collect and whenever extra evidence becomes available.

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Table: A simple dataset for MENINGITIS diagnosis with descriptive features that describe the presence or absence of three common symptoms of the disease: HEADACHE, FEVER, and VOMITING.

ID	HEADACHE	FEVER	VOMITING	MENINGITIS
1	true	true	false	false
2	false	true	false	false
3	true	false	true	false
4	true	false	true	false
5	false	true	false	true
6	true	false	true	false
7	true	false	true	false
8	true	false	true	true
9	false	true	false	false
10	true	false	true	true

- Each feature in a dataset is a **random variable**, and the **sample space** for the domain associated with a prediction problem is the set of all possible combinations of assignments of values to features.
- An **experiment**: a row in a dataset; it maps a set of descriptive feature values to a target feature value. **Event**: assignment of a set of descriptive features with values.
- A **probability function**, $P()$, returns the probability of a feature taking a specific value. $P(\text{FEVER} = \text{true})$?
- A **joint probability**: probability of an assignment of specific values to multiple different features. $P(\text{MENINGITIS} = \text{true}, \text{HEADACHE} = \text{true})$?
- A **conditional probability**: probability of one feature taking a specific value given that we already know the value of a different feature. $P(\text{MENINGITIS} = \text{true} \mid \text{HEADACHE} = \text{true})$?

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$$\mathbf{P}(H, F, V, M) = \begin{bmatrix} P(h, f, v, m), & P(\neg h, f, v, m) \\ P(h, f, v, \neg m), & P(\neg h, f, v, \neg m) \\ P(h, f, \neg v, m), & P(\neg h, f, \neg v, m) \\ P(h, f, \neg v, \neg m), & P(\neg h, f, \neg v, \neg m) \\ P(h, \neg f, v, m), & P(\neg h, \neg f, v, m) \\ P(h, \neg f, v, \neg m), & P(\neg h, \neg f, v, \neg m) \\ P(h, \neg f, \neg v, m), & P(\neg h, \neg f, \neg v, m) \\ P(h, \neg f, \neg v, \neg m), & P(\neg h, \neg f, \neg v, \neg m) \end{bmatrix}$$

- Given a joint probability distribution, we can compute the probability of any event in the domain that it covers by summing over the cells in the distribution where that event is true.
- Example: compute the probability of $P(h)$ in the domain specified by the joint probability distribution $\mathbf{P}(H, F, V, M)$.

- out.**

Probability: Frequentist vs. Bayesian Perspective

Frequentist Perspective:

- Probability represents the long-run frequency of an event occurring in repeated, identical experiments.
- Focuses on objective, observable outcomes and considers probabilities as the limit of frequencies.
- Key concept: Law of Large Numbers.

Probability: Frequentist vs. Bayesian Perspective

Frequentist Perspective:

- Probability represents the long-run frequency of an event occurring in repeated, identical experiments.
- Focuses on objective, observable outcomes and considers probabilities as the limit of frequencies.
- Key concept: Law of Large Numbers.

Bayesian Perspective:

- Probability is a measure of belief or uncertainty about the likelihood of an event.
- Incorporates prior knowledge (prior probability) and updates beliefs based on new evidence (posterior probability) using Bayes' Theorem.
- Emphasizes the subjective nature of probability.

Probability: Frequentist vs. Bayesian Perspective

Frequentist

Objective, based on observed frequencies.

Frequentist: $\frac{\# \text{occurrences}}{\# \text{total}} \rightarrow P(A)$

Bayesian:
$$\underbrace{P(A | B)}_{\text{posterior}} = \frac{\overbrace{P(B | A)}^{\text{likelihood}} \overbrace{P(A)}^{\text{prior}}}{P(B)}$$

Bayesian

$P(\text{Event} \mid \text{Evidence})$: Subjective, incorporates prior beliefs and updates with new evidence.

Bayes' Theorem

$$P(X|Y) = \frac{P(Y|X)P(X)}{P(Y)}$$

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Bayes' Theorem

$$P(d|t) = \frac{P(t|d)P(d)}{P(t)}$$

$$P(d|t) = \frac{P(t|d)P(d)}{P(t)}$$

Using the **Theorem of Total Probability**:

$$\begin{aligned} P(t) &= P(t|d)P(d) + P(t|\neg d)P(\neg d) \\ &= (0.99 \times 0.0001) + (0.01 \times 0.9999) = 0.0101 \end{aligned}$$

$$\begin{aligned} P(d|t) &= \frac{0.99 \times 0.0001}{0.0101} \\ &= 0.0098 \end{aligned}$$

Deriving Bayes theorem:

$$P(X, Y) = P(X, Y)$$

The theorem is easily derived from the **product rule** which defines $P(X \wedge Y)$ (i.e. also $P(Y \wedge X)$):

$$P(Y|X)P(X) = P(X|Y)P(Y)$$

$$\frac{P(X|Y)P(Y)}{P(Y)} = \frac{P(Y|X)P(X)}{P(Y)}$$

$$\frac{\cancel{P(X|Y)}\cancel{P(Y)}}{\cancel{P(Y)}} = \frac{P(Y|X)P(X)}{P(Y)}$$

$$\Rightarrow P(X|Y) = \frac{P(Y|X)P(X)}{P(Y)}$$

- The divisor is the prior probability of the evidence
- This division functions as a normalization mechanism (actually, it is constant for all X_i).

$$0 \leq P(X|Y) \leq 1$$

$$\sum_i P(X_i|Y) = 1.0$$

- We can calculate this divisor directly from the dataset.

$$P(Y) = \frac{|\{\text{rows where } Y \text{ is the case}\}|}{|\{\text{rows in the dataset}\}|}$$

- Or, we can use the **Theorem of Total Probability** to calculate this divisor.

$$P(Y) = \sum_i P(Y|X_i)P(X_i) \quad (1)$$

- To make Bayesian predictions, we generate the probability of the event that a target feature, t , takes a specific level, l , given the assignment of values to a set of descriptive features, q , from a query instance.
- With this in mind, the definition of Bayes' Theorem can be generalized so that it can take into account more than one piece of evidence (each descriptive feature value is a separate piece of evidence).

Generalized Bayes' Theorem

$$P(t = l | \mathbf{q}[1], \dots, \mathbf{q}[m]) = \frac{P(\mathbf{q}[1], \dots, \mathbf{q}[m] | t = l) P(t = l)}{P(\mathbf{q}[1], \dots, \mathbf{q}[m])}$$

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$$P(\mathbf{q}[1], \dots, \mathbf{q}[m]) = P(\mathbf{q}[1]) \times P(\mathbf{q}[2]|\mathbf{q}[1]) \times \dots \times P(\mathbf{q}[m]|\mathbf{q}[m-1], \dots, \mathbf{q}[2], \mathbf{q}[1])$$

- To apply the chain rule to a conditional probability we just add the conditioning term to each term in the expression:

$$P(\mathbf{q}[1], \dots, \mathbf{q}[m] | t = l) = \\ P(\mathbf{q}[1] | t = l) \times P(\mathbf{q}[2] | \mathbf{q}[1], t = l) \times \dots \\ \dots \times P(\mathbf{q}[m] | \mathbf{q}[m-1], \dots, \mathbf{q}[3], \mathbf{q}[2], \mathbf{q}[1], t = l)$$

Bayesian Prediction

ID	HEADACHE	FEVER	VOMITING	MENINGITIS
1	true	true	false	false
2	false	true	false	false
3	true	false	true	false
4	true	false	true	false
5	false	true	false	true
6	true	false	true	false
7	true	false	true	false
8	true	false	true	true
9	false	true	false	false
10	true	false	true	true

HEADACHE	FEVER	VOMITING	MENINGITIS
true	false	true	?

$$P(M|h, \neg f, v) = ?$$

- In the terms of Bayes' Theorem this problem can be stated as:

$$P(M|h, \neg f, v) = \frac{P(h, \neg f, v|M) \times P(M)}{P(h, \neg f, v)}$$

- There are two values in the domain of the MENINGITIS feature, 'true' and 'false', so we have to do this calculation twice.

- We will do the calculation for m first
- To carry out this calculation we need to know the following probabilities: $P(m)$, $P(h, \neg f, v)$ and $P(h, \neg f, v \mid m)$.

ID	HEADACHE	FEVER	VOMITING	MENINGITIS
1	true	true	false	false
2	false	true	false	false
3	true	false	true	false
4	true	false	true	false
5	false	true	false	true
6	true	false	true	false
7	true	false	true	false
8	true	false	true	true
9	false	true	false	false
10	true	false	true	true

- We can calculate $P(h, \neg f, v \mid m)$ from the table. However, as an exercise, we will use the chain rule to calculate it:

$$P(h, \neg f, v \mid m) = ?$$

ID	HEADACHE	FEVER	VOMITING	MENINGITIS
1	true	true	false	false
2	false	true	false	false
3	true	false	true	false
4	true	false	true	false
5	false	true	false	true
6	true	false	true	false
7	true	false	true	false
8	true	false	true	true
9	false	true	false	false
10	true	false	true	true

- Using the chain rule calculate:

$$\begin{aligned} P(h, \neg f, v \mid m) &= P(h \mid m) \times P(\neg f \mid h, m) \times P(v \mid \neg f, h, m) \\ &= \frac{|\{\mathbf{d}_8, \mathbf{d}_{10}\}|}{|\{\mathbf{d}_5, \mathbf{d}_8, \mathbf{d}_{10}\}|} \times \frac{|\{\mathbf{d}_8, \mathbf{d}_{10}\}|}{|\{\mathbf{d}_8, \mathbf{d}_{10}\}|} \times \frac{|\{\mathbf{d}_8, \mathbf{d}_{10}\}|}{|\{\mathbf{d}_8, \mathbf{d}_{10}\}|} \\ &= \frac{2}{3} \times \frac{2}{2} \times \frac{2}{2} = 0.6666 \end{aligned}$$

- So the calculation of $P(m|h, \neg f, v)$ is:

$$\begin{aligned}
 P(m|h, \neg f, v) &= \frac{\left(P(h|m) \times P(\neg f|h, m) \right. \\
 &\quad \left. \times P(v|\neg f, h, m) \times P(m) \right)}{P(h, \neg f, v)} \\
 &= \frac{0.6666 \times 0.3}{0.6} = 0.3333
 \end{aligned}$$

- The corresponding calculation for $P(\neg m|h, \neg f, v)$ is:

$$\begin{aligned} P(\neg m \mid h, \neg f, v) &= \frac{P(h, \neg f, v \mid \neg m) \times P(\neg m)}{P(h, \neg f, v)} \\ &= \frac{\left(P(h \mid \neg m) \times P(\neg f \mid h, \neg m) \right. \\ &\quad \left. \times P(v \mid \neg f, h, \neg m) \times P(\neg m) \right)}{P(h, \neg f, v)} \\ &= \frac{0.7143 \times 0.8 \times 1.0 \times 0.7}{0.6} = 0.6667 \end{aligned}$$

$$P(\neg m|h, \neg f, v) = 0.6667$$

- These calculations tell us that it is twice as probable that the patient does not have meningitis than it is that he/she does even though the patient is suffering from a headache and is vomiting!
- In this case, although the likelihood of suffering from a headache and vomiting is quite high when someone has meningitis, the prior probability of having meningitis is quite low.
- So, even when we take the evidence into account, the posterior probability of having meningitis remains low.
- This can seem counter-intuitive at first.

The Paradox of the False Positive

- The mistake of forgetting to factor in the prior gives rise to the **paradox of the false positive** which states that in order to make predictions about a rare event, the model has to be as accurate as the prior of the event is rare; otherwise, there is a significant chance of **false positives** predictions (i.e., predicting the event when it is not the case).
- In other words, the paradox of the false positive highlights the importance of considering both the accuracy of a test and the prior probability of the event being tested for when interpreting test results.

- Calculating exact probabilities for each of the possible target levels is often very useful to a human decision maker, for example, a doctor.
- However, if we are trying to build a predictive model that automatically assigns a target level to a query instance, then we need to decide how the model will make a prediction based on the computed probabilities.
- Approach: have the model return the target level that has the highest posterior probability given the state of the descriptive features in the query.
- A prediction model that works in this way is making a **maximum a posteriori (MAP)** prediction.

Bayesian MAP Prediction Model

$$\begin{aligned} \mathbb{M}_{MAP}(\mathbf{q}) &= \underset{l \in levels(t)}{\operatorname{argmax}} P(t = l \mid \mathbf{q}[1], \dots, \mathbf{q}[m]) \\ &= \underset{l \in levels(t)}{\operatorname{argmax}} \frac{P(\mathbf{q}[1], \dots, \mathbf{q}[m] \mid t = l) \times P(t = l)}{P(\mathbf{q}[1], \dots, \mathbf{q}[m])} \end{aligned}$$

Bayesian MAP Prediction Model (without normalization)

$$\mathbb{M}_{MAP}(\mathbf{q}) = \underset{l \in \text{levels}(t)}{\operatorname{argmax}} P(\mathbf{q}[1], \dots, \mathbf{q}[m] \mid t = l) \times P(t = l)$$

$$\begin{aligned} \mathbb{M}_{MAP}(\mathbf{q}) &= \underset{l \in \text{levels}(t)}{\operatorname{argmax}} P(t = l \mid \mathbf{q}[1], \dots, \mathbf{q}[m]) \\ &= \underset{l \in \text{levels}(t)}{\operatorname{argmax}} \frac{P(\mathbf{q}[1], \dots, \mathbf{q}[m] \mid t = l) \times P(t = l)}{P(\mathbf{q}[1], \dots, \mathbf{q}[m])} \end{aligned}$$

1. *Journal of the American Medical Association*, 2000; 284: 2689-2695.

$$\mathbb{M}_{MAP}(\mathbf{q}) = \underset{l \in \text{levels}(t)}{\operatorname{argmax}} P(\mathbf{q}[1], \dots, \mathbf{q}[m] \mid t = l) \times P(t = l)$$

Unfortunately, we are not done yet! There is still a problem, illustrated through the following example.

Consider the following dataset:

ID	HEADACHE	FEVER	VOMITING	MENINGITIS
1	true	true	false	false
2	false	true	false	false
3	true	false	true	false
4	true	false	true	false
5	false	true	false	true
6	true	false	true	false
7	true	false	true	false
8	true	false	true	true
9	false	true	false	false
10	true	false	true	true

And consider the query:

HEADACHE	FEVER	VOMITING	MENINGITIS
true	true	false	?

ID	HEADACHE	FEVER	VOMITING	MENINGITIS
1	true	true	false	false
2	false	true	false	false
3	true	false	true	false
4	true	false	true	false
5	false	true	false	true
6	true	false	true	false
7	true	false	true	false
8	true	false	true	true
9	false	true	false	false
10	true	false	true	true

$$P(m \mid h, f, \neg v) = ?$$

$$P(\neg m \mid h, f, \neg v) = ?$$

$$\begin{aligned}
 P(m \mid h, f, \neg v) &= \frac{\left(P(h \mid m) \times P(f \mid h, m) \right. \\
 &\quad \left. \times P(\neg v \mid f, h, m) \times P(m) \right)}{P(h, f, \neg v)} \\
 &= \frac{0.6666 \times 0 \times 0 \times 0.3}{0.1} = 0
 \end{aligned}$$

$$\begin{aligned}
 P(\neg m \mid h, f, \neg v) &= \frac{\left(P(h \mid \neg m) \times P(f \mid h, \neg m) \right. \\
 &\quad \left. \times P(\neg v \mid f, h, \neg m) \times P(\neg m) \right)}{P(h, f, \neg v)} \\
 &= \frac{0.7143 \times 0.2 \times 1.0 \times 0.7}{0.1} = 1.0
 \end{aligned}$$

Curse of Dimensionality

- As the number of descriptive features grows, the number of potential conditioning events grows.
- It is crucial to ensure enough instances in the training dataset matching these conditions, making the probability calculation possible.
- this is due to **data fragmentation** (an instance of the **curse of dimensionality**).
- This necessitates an **exponential increase** in the size of the dataset for each newly added descriptive feature.

- If knowledge of one event has no effect on the probability of another event, and *vice versa*, then the two events are **independent** of each other.
- If two events X and Y are independent then:

$$P(X|Y) = P(X)$$

$$P(X, Y) = P(X) \times P(Y)$$

- Recall, that when two event are dependent these rules are:

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

$$P(X, Y) = P(X|Y) \times P(Y) = P(Y|X) \times P(X)$$

Conditional Independence and Factorization

- Full independence between events is quite rare.
- A more common phenomenon is that two, or more, events may be independent if we know that a third event has happened.
- This is known as **conditional independence**.

- For two events, X and Y , that are conditionally independent given knowledge of a third event, here Z , the definition of the probability of a joint event and conditional probability are:

$$P(X|Y, Z) = P(X|Z)$$

$$P(X, Y|Z) = P(X|Z) \times P(Y|Z)$$

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

$$\begin{aligned} P(X, Y) &= P(X|Y) \times P(Y) \\ &= P(Y|X) \times P(X) \end{aligned}$$

X and Y are dependent

$$P(X|Y) = P(X)$$

$$P(X, Y) = P(X) \times P(Y)$$

X and Y are independent

- If the event $t = I$ causes the events $\mathbf{q}[1], \dots, \mathbf{q}[m]$ to happen then the events $\mathbf{q}[1], \dots, \mathbf{q}[m]$ are conditionally independent of each other given knowledge of $t = I$ and the chain rule definition can be simplified as follows:

Conditional Independence and Factorization

- Using this we can simplify the calculations in Bayes' Theorem, under the assumption of conditional independence between the descriptive features given the level l of the target feature:

$$P(t = l \mid \mathbf{q}[1], \dots, \mathbf{q}[m]) = \frac{\left(\prod_{i=1}^m P(\mathbf{q}[i] \mid t = l) \right) \times P(t = l)}{P(\mathbf{q}[1], \dots, \mathbf{q}[m])}$$

- The joint probability distribution for the meningitis dataset.

$$\mathbf{P}(H, F, V, M) = \begin{bmatrix} P(h, f, v, m), & P(\neg h, f, v, m) \\ P(h, f, v, \neg m), & P(\neg h, f, v, \neg m) \\ P(h, f, \neg v, m), & P(\neg h, f, \neg v, m) \\ P(h, f, \neg v, \neg m), & P(\neg h, f, \neg v, \neg m) \\ P(h, \neg f, v, m), & P(\neg h, \neg f, v, m) \\ P(h, \neg f, v, \neg m), & P(\neg h, \neg f, v, \neg m) \\ P(h, \neg f, \neg v, m), & P(\neg h, \neg f, \neg v, m) \\ P(h, \neg f, \neg v, \neg m), & P(\neg h, \neg f, \neg v, \neg m) \end{bmatrix}$$

ID	HEADACHE	FEVER	VOMITING	MENINGITIS
1	true	true	false	false
2	false	true	false	false
3	true	false	true	false
4	true	false	true	false
5	false	true	false	true
6	true	false	true	false
7	true	false	true	false
8	true	false	true	true
9	false	true	false	false
10	true	false	true	true

- Calculate the factors from the data.

$$Factor_1 : < P(M) >$$

$$Factor_2 : < P(h|m), P(h|\neg m) >$$

$$Factor_3 : < P(f|m), P(f|\neg m) >$$

$$Factor_4 : < P(v|m), P(v|\neg m) >$$

$$Factor_1 : < P(m) = 0.3 >$$

$$Factor_2 : < P(h|m) = 0.6666, P(h|\neg m) = 0.7413 >$$

$$Factor_3 : < P(f|m) = 0.3333, P(f|\neg m) = 0.4286 >$$

$$Factor_4 : < P(v|m) = 0.6666, P(v|\neg m) = 0.5714 >$$

$$P(m|h, f, \neg v) = \frac{P(h|m) \times P(f|m) \times P(\neg v|m) \times P(m)}{\sum_i P(h|M_i) \times P(f|M_i) \times P(\neg v|M_i) \times P(M_i)} =$$
$$\frac{0.6666 \times 0.3333 \times 0.3333 \times 0.3}{(0.6666 \times 0.3333 \times 0.3333 \times 0.3) + (0.7143 \times 0.4286 \times 0.4286 \times 0.7)} = 0.1948$$

$$P(m|h, f, \neg v) = 0.1948$$

$$P(\neg m|h, f, \neg v) = 0.8052$$

- As before, the MAP prediction would be
MENINGITIS = *'false'*
- The posterior probabilities are not as extreme!

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- ## Naive Bayes Classifier

$$\mathbb{M}(\mathbf{q}) = \underset{l \in \text{levels}(t)}{\operatorname{argmax}} \left(\prod_{i=1}^m P(\mathbf{q}[i] \mid t = l) \right) \times P(t = l)$$

Naive Bayes is simple to train!

- 1 calculate the priors for each of the target levels
- 2 calculate the conditional probabilities for each feature given each target level.

Table: A dataset from a loan application fraud detection domain.

ID	CREDIT HISTORY	GUARANTOR/ COAPPLICANT	ACCOMODATION	FRAUD
1	current	none	own	true
2	paid	none	own	false
3	paid	none	own	false
4	paid	guarantor	rent	true
5	arrears	none	own	false
6	arrears	none	own	true
7	current	none	own	false
8	arrears	none	own	false
9	current	none	rent	false
10	none	none	own	true
11	current	coapplicant	own	false
12	current	none	own	true
13	current	none	rent	true
14	paid	none	own	false
15	arrears	none	own	false
16	current	none	own	false
17	arrears	coapplicant	rent	false
18	arrears	none	free	false
19	arrears	none	own	false
20	paid	none	own	false

A Worked Example

- There are two levels in the target feature domain, four levels in the CREDIT HISTORY domain, three in the GUARANTOR/COAPPLICANT domain, and three in the ACCOMMODATION domain.
- So we need to calculate $2 + (4 \times 2) + (3 \times 2) + (3 \times 2) = 22$ probabilities.
- This sounds like a lot of probabilities considering the size of the example dataset, but these 22 probabilities (see next slide) would suffice no matter how many new instances are added to the dataset (100s, 1000s, or even millions).
- This is an example of the compactness of a naive Bayes representation.

$P(fr) = 0.3$	$P(\neg fr) = 0.7$
$P(CH = 'none' fr) = 0.1666$	$P(CH = 'none' \neg fr) = 0$
$P(CH = 'paid' fr) = 0.1666$	$P(CH = 'paid' \neg fr) = 0.2857$
$P(CH = 'current' fr) = 0.5$	$P(CH = 'current' \neg fr) = 0.2857$
$P(CH = 'arrears' fr) = 0.1666$	$P(CH = 'arrears' \neg fr) = 0.4286$
$P(GC = 'none' fr) = 0.8334$	$P(GC = 'none' \neg fr) = 0.8571$
$P(GC = 'guarantor' fr) = 0.1666$	$P(GC = 'guarantor' \neg fr) = 0$
$P(GC = 'coapplicant' fr) = 0$	$P(GC = 'coapplicant' \neg fr) = 0.1429$
$P(ACC = 'own' fr) = 0.6666$	$P(ACC = 'own' \neg fr) = 0.7857$
$P(ACC = 'rent' fr) = 0.3333$	$P(ACC = 'rent' \neg fr) = 0.1429$
$P(ACC = 'free' fr) = 0$	$P(ACC = 'free' \neg fr) = 0.0714$

Table: The probabilities needed by a Naive Bayes prediction model calculated from the dataset. Notation key: FR=FRAUDULENT, CH=CREDIT HISTORY, GC = GUARANTOR/COAPPLICANT, ACC = ACCOMODATION, T='true', F='false'.

$P(fr) = 0.3$	$P(\neg fr) = 0.7$
$P(CH = 'none' fr) = 0.1666$	$P(CH = 'none' \neg fr) = 0$
$P(CH = 'paid' fr) = 0.1666$	$P(CH = 'paid' \neg fr) = 0.2857$
$P(CH = 'current' fr) = 0.5$	$P(CH = 'current' \neg fr) = 0.2857$
$P(CH = 'arrears' fr) = 0.1666$	$P(CH = 'arrears' \neg fr) = 0.4286$
$P(GC = 'none' fr) = 0.8334$	$P(GC = 'none' \neg fr) = 0.8571$
$P(GC = 'guarantor' fr) = 0.1666$	$P(GC = 'guarantor' \neg fr) = 0$
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$P(ACC = 'own' fr) = 0.6666$	$P(ACC = 'own' \neg fr) = 0.7857$
$P(ACC = 'rent' fr) = 0.3333$	$P(ACC = 'rent' \neg fr) = 0.1429$
$P(ACC = 'free' fr) = 0$	$P(ACC = 'free' \neg fr) = 0.0714$

CREDIT HISTORY	GUARANTOR/COAPPLICANT	ACCOMODATION	FRAUDULENT
paid	none	rent	?

1. *Journal of Management Studies*, 1997, 34, 1, 1-14.

1. *Journal of Management Studies*, 1997, 34, 1, 1-14.

$$P(\text{ACC} = \text{'rent'} \mid \neg fr) = 0.1429$$

$$\left(\prod_{k=1}^m P(\mathbf{q}[k] \mid \neg fr) \right) \times P(\neg fr) = 0.0245$$

The model is generalizing beyond the dataset!

ID	CREDIT HISTORY	GUARANTOR/ COAPPLICANT	ACCOMMODATION	FRAUD
1	current	none	own	true
2	paid	none	own	false
3	paid	none	own	false
4	paid	guarantor	rent	true
5	arrears	none	own	false
6	arrears	none	own	true
7	current	none	own	false
8	arrears	none	own	false
9	current	none	rent	false
10	none	none	own	true
11	current	coapplicant	own	false
12	current	none	own	true
13	current	none	rent	true
14	paid	none	own	false
15	arrears	none	own	false
16	current	none	own	false
17	arrears	coapplicant	rent	false
18	arrears	none	free	false
19	arrears	none	own	false
20	paid	none	own	false

CREDIT HISTORY	GUARANTOR/COAPPLICANT	ACCOMMODATION	FRAUDULENT
paid	none	rent	<i>'false'</i>

- The relative ranking of the likelihood of the target levels is, to a certain extent, robust to errors in the calculation of the exact probabilities.
- The naive Bayes model is relatively robust to data fragmentation and the curse of dimensionality. (Particularly important with small datasets or with sparse data.)
- One application domain where sparse data is the norm is in text analytics (e.g., spam filtering), and where naive Bayes models are often successful.

- The naive Bayes model can easily be adapted to handle missing feature values by dropping from the product of the evidence events the conditional probabilities that specify features with values not in the data. This may have a negative effect on the accuracy of posterior probabilities and may not translate directly into prediction errors.
- Another advantage is the compactness of the naive Bayes model with which a very large dataset can be represented.

- The naive Bayes model can easily be adapted to handle missing feature values by dropping from the product of the evidence events the conditional probabilities that specify features with values not in the data. This may have a negative effect on the accuracy of posterior probabilities and may not translate directly into prediction errors.
- Another advantage is the compactness of the naive Bayes model with which a very large dataset can be represented.
- Naive Bayes models may not be as powerful as some other prediction models, but they often provide reasonable accuracy results for prediction tasks with categorical targets, while being robust to the curse of dimensionality and also being easy to train.

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(2)

- A Naive Bayes classifier naively assumes that each of the descriptive features in a domain is conditionally independent of all of the other descriptive features, given the state of the target feature.

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- A Naive Bayes classifier naively assumes that each of the descriptive features in a domain is conditionally independent of all of the other descriptive features, given the state of the target feature.
- This assumption, although often wrong, enables the Naive Bayes model to maximally factorise the representation that it uses of the domain.

$$P(t|\mathbf{d}) = \frac{P(\mathbf{d}|t) \times P(t)}{P(\mathbf{d})} \quad (2)$$

- A Naive Bayes classifier naively assumes that each of the descriptive features in a domain is conditionally independent of all of the other descriptive features, given the state of the target feature.
- This assumption, although often wrong, enables the Naive Bayes model to maximally factorise the representation that it uses of the domain.
- Surprisingly, given the naivety and strength of the assumption it depends upon, a Naive Bayes model often performs reasonably well.

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The slides used in this course are based on the official textbook materials, with modifications made where necessary to suit the course requirements and enhance the learning experience.